A rate-[multi-utility] theorem

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Here's a version of the rate-distortion theorem with many utility functions.

Theorem 1. *Fix k utility functions $u_1, u_2, \ldots, u_k \colon \mathcal{X} \times \mathcal{Z} \to \mathbb{R}$. Fix an input random variable X, i.e., with domain $\mathcal{X}$. For a noisy channel with capacity r used at rate r, a tuple of values for these k expected utilities is attainable when coding iff it is attainable by some single-letter channel. That is, a tuple of expected average utilities is attainable when coding input sequences $x^{1:n}$ to output sequences $z^{1:n}$ iff there is a joint distribution on X and Z such that $I(X;Z) < rc$ with these expected utilities.*

todo: one might need to hedge the above statement a tiny bit to make the proof apply (and also hedge some statements in the proof sketch below, actually, which is certainly careless about e.g. S being compact atm) — i think one might lose an epsilon in one direction

Proof. If a tuple of expected average utilities is attainable by a coding scheme, then one can use the same index trick as in the proof of vanilla rate-distortion to get a test channel with the same tuple of expected utilities that satisfies the desired bound on the mutual information.

It remains to show the converse — that if a utility tuple is attainable by a test channel, then there is a coding scheme achieving it as well. Let $S \subseteq \mathbb{R}^k$ be the set of expected average utility tuples attainable from such codings. S is convex (this is because one can take $n = n_1 + n_2$ and use one coding method for the batch made of the first n_1 indices and another coding method for the batch made of the remaining n_2 indices). Let $S' \subseteq \mathbb{R}^k$ be the set of expected utility tuples attainable from such test channels. For an arbitrary linear combination of utilities, the usual rate-distortion theorem tells us that it is attainable iff it is attainable by a test channel. This means that in every linear direction, the projection of S covers the projection of S'. Since S is convex, this implies that $S' \subseteq S$ (if not, one could find a point of S' outside of S, take the line from it to the closest point in S, and project to that line, getting a shadow of S' which is not covered by a shadow of S). $\square$